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# Particle swarm optimization

## Pseudo code

```
Initialize particles
for each particle
{
    initialize position (x)
    initialize velocity (v)
    calculate fitness

    pbest = x
}

gbest = best pbest

for generation in generations
{
    for each particle
    {
        r1 = random (0, 1)
        r2 = random (0, 1)
        v = w * v + c1 * r1 * (pbest - x)
                  + c2 * r2 * (gbest - x)

        x = x + v

        calculate fitness (x)

        if fitness (x) > fitness (pbest)
            pbest = x

        if fitness (x) > fitness (gbest)
```

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gbest = x

}

}

print (gbest)

end

proceed.

Neural N/w optimization

Title: Optimized particle swarm optimization & its application to artificial neural network training

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The paper focuses on improving the training of artificial neural networks using an optimized version of particle swarm optimization

Traditional neural network training generally uses back propagation. However, training can face issues such as local minima & dependence on initial weights. The paper investigates PSO as a population-based optimization method for finding better neural network parameters.

Code Summary

The code implements particle swarm optimization to optimize the weight & biases of a small artificial neural network for solving the XOR problem. It initializes 30 particles, where each particle represents a possible set of 13 neural-network parameters. The neural network uses a Sigmoid activation function, & mean squared error is used as the objective function. During 200 iterations, the particles update their velocities & positions based on their personal best & the global best solutions. The best set weights found by PSO is then used to make the final predictions.

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## Output Summary

The output shows the optimization progress by displaying the best MSE at different iterations. As the iterations increase, the MSE generally decreases, showing that PSO is improving the neural network parameters. At the end, the program displays the final MSE & the neural network's prediction for the four XOR inputs. Predictions close to 0 for the actual output 0 & close to 1 for actual output 1 indicate that the optimized neural network has successfully learned the XOR problem.

Coding completed

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